

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
---------------------	--	---------------------	-------

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) Question Count: 5	Total Marks: 25
--	------------------------

Code of Conduct

I M.V.RAKESH REDDY/192524164 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.V.Rakesh Reddy

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

1) Define cloud computing and Explain

Defination:

Cloud computing is the on-demand delivery of computing services - including servers, storage, databases, networking, software, analytics, and intelligence - over the internet ("the cloud") with pay-as-you-go pricing. It allows users to access and use its resources without owning or maintaining physical infrastructure.

Four key characteristics

1. On demand self-service

Users can provision computing resources automatically without requiring human interaction with service providers.

2. Broad Network Access

Resources are available over the network and accessed through standard mechanisms (web browsers, mobile, app, APIs).

3. Resource Pooling

Providers, computing resources, are pooled to serve multiple customers using a multi-tenement model.

4. Rapid Elasticity

- Resources can be scaled up or down quickly often automatically.
- To the consumer available resource appear unlimited and can be provided in any quantity at any time.

5. Measured service.

Cloud systems automatically control and optimize resource usage through metering usage is monitored controlled and reported (Pay as you go model)

3. Resource Pooling

Providers, computing resources, are pooled to serve multiple customers using a multi-tenement model.

4. Rapid Elasticity

- Resources can be scaled up or down quickly often automatically.
- To the consumer available resource appear unlimited and can be provided in any quantity at any time.

5. Measured service.

Cloud systems automatically control and optimize resource usage through metering usage is monitored controlled and reported (Pay as you go model)

3. Resource Pooling

Providers, computing resources, are pooled to serve multiple customers using a multi-tenement model.

4. Rapid Elasticity

- Resources can be scaled up or down quickly often automatically.
- To the consumer available resource appear unlimited and can be provided in any quantity at any time.

5. Measured service.

Cloud systems automatically control and optimize resource usage through metering usage is monitored controlled and reported (Pay as you go model)

2011.

IBM.

Enterprise

are

2015

AI, Edge.

Next Generation.

3) Role of Server Virtualization in Cloud Computing

Server virtualization is a technology that divides a single physical server into multiple isolated virtual machine (VMs) each running its own operating system and application.

How it work:

Physical server

|- Hypervision (Vmware, xen, kvm)

|- VM1 (Windows, APP A)

VM2 (Linux, APP B)

VM3 (Linux, APP C)

VM4 (Windows, APP C)

4) Role of Data centers in delivering cloud services.

A Data center is a physical facility that houses computers, systems storage, systems, networking equipment and related infrastructure for storing, processing and managing data.

Component

Function

Servers

Computing Power

Storage

Data storage

Networking

switches, routers, load

Power Systems

UPS, generators, power distribution

Cooling

AC units, cooling systems

Security

Physical and Cybersecurity measures.

Role in. Cloud services.

1. Physical foundation.
2. Redundancy and Reliability
3. Global reach
4. Economics of scale
5. security

5) Differentiate IaaS, PaaS, SaaS, and XaaS

Real world Example:

1. IaaS - AWS

- Ex: Launching a virtual server on AWS EC2
- User gets: Virtual machine CPU RAM.
- User configuration: operating system, security applications
- Provides, manages: Physical hardware.

2. PaaS - Google App Engine.

Example: Deploying a Python/Java

User gets: Development platform, and runtime

User configures: Applications, code only.

User manages: only applications, and data

Provider manages: OS, runtime.

3. SaaS. Salesforce CRM.

Example: using Salesforce, pay customore.

User gets: Complete CRM application

User configures: settings

User manages: Nothing technical

Provides: Everything

4. XaaS

Example: ~~YaaS~~ Flight Reservation

User gets: Cloud based application platform

Service: Custom application

Provider manages: Everything while user

focuses on building

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow